

Question 1: A 50 kW carrier is to be modulated to a level of (i) 80% (ii) 10%. What is the total sideband power in each case?

Ans:-

carrier power $P_c = 50 \text{ kW}$

(i) modulation $m = 80\% = 0.8$

Sideband Power $P_{sb} = \frac{1}{2} m^2 P_c$

$$= \frac{1}{2} \times (0.8)^2 \times 50$$

$$= 16 \text{ kW}$$

Ans

(ii) modulation $m = 10\%$
 $= 0.1$

$$\therefore \text{Sideband Power} = \frac{1}{2} \times (0.1)^2 \times 50$$

$$= 0.25 \text{ kW}$$

Question 2 An AM wave is represented by the expression $V = (10 + 6 \cos 4356t) \sin 133 \times 10^3 t$ volts. Relate this with standard AM wave equation and find values of each variable.

Ans: Given

$$V = (10 + 6 \cos 4356t) \sin 133 \times 10^3 t$$

standard equation

$$V = E_c (1 + m \cos \omega_s t + \sin \omega_c t) \dots \textcircled{i}$$

$$\therefore V = 10 (1 + \frac{6}{10} \cos 4356t) \sin 133 \times 10^3 t \dots \textcircled{ii}$$

Comparing $\textcircled{i}$ & $\textcircled{ii}$

$$E_c = 10 \text{ V}$$

$$m = \frac{6}{10} = 0.6$$

$$\omega_s = 4356 \text{ rad/s}$$

$$\omega_c = 133 \times 10^3 \text{ rad/s}$$